



Week 3 Milestone Worksheet

Instructions: Provide responses to all items in the boxes. The worksheet consists of Sections A–E. Work on this worksheet one section at a time throughout your week, and return to Savanna after each section for the next set of content and further instructions.

SECTION A: Problem Statement

Step 1: Your GCGO

Which Grand Challenge or Great Opportunity (GCGO) do you want to play a part in addressing? (Pick one.)

As a reminder, the GCGOs are:

- Urbanization
- Education
- Infrastructure
- Healthcare
- Climate change
- Governance
- Job creation
- Agriculture
- Natural resources
- Arts, culture, and design
- Tourism
- Empowerment of women
- Regional integration
- Wildlife conservation

GCGO:

Governance



Step 2: Describe Your Problem

You are going to take a first pass at briefly describing your chosen problem. This can be any problem that speaks to you, as long as it is a real-life occurrence that is clearly linked to your chosen GCGO, that occurs in a certain place and for certain people (and/or animals), and that it can be clearly defined.

For example, if you chose wildlife conservation as your GCGO, you might first state your chosen problem as:

There are very few white rhinos left in Kenya, and they are in danger of becoming extinct.

Another example is if you choose infrastructure as your GCGO:

Residents of major cities in South Africa endure prolonged periods without electricity, significantly hampering their ability to generate income.

Note that this is just your first attempt at stating the problem, and you don't need to quantify it yet. In order to get to your official problem statement (which does need to be quantifiable), first answer the following questions. The more specific your answers, the better. You may also ask Google, Wikipedia, ChatGPT, and/or other reliable online sources to help you. Please be sure to cite (give credit to) any sources that you use.

Describe your problem using **What/Who/When/Where/Why/How....**

1. **What** is the problem? What is reality like because of this problem? What will the reality be like if the problem continues?

The problem is the increasing number of cyberattacks and online fraud affecting individuals and organizations. Many people and businesses do not have strong cybersecurity protection. Because of this, hackers can steal

personal information, passwords, and money. If this problem continues, more people and organizations will lose sensitive data and trust in digital systems.

2. **Who** does this problem impact, directly and indirectly? Who contributes to the problem?

This problem affects internet users, businesses, banks, and government institutions. Individuals may lose money through online scams, while organizations may lose important data. Cybercriminals and hackers contribute to the problem by exploiting weak security systems.

3. **When** did this problem begin? When does it occur?

The problem began increasing as internet usage and digital technology became more common. It occurs whenever people use online services such as social media, online banking, email, or digital platforms.

4. **Where** is this problem occurring? What is the context in which it occurs?

This problem occurs worldwide, including in Kenya. It mainly happens in digital environments such as websites, mobile applications, email systems, and online financial platforms.

5. **Why** is this a problem? What are the pain points or gaps? Why do you personally care about this problem?

Cybercrime leads to financial losses, identity theft, and loss of important information. Many people are not aware of cybersecurity risks, which makes them easy targets for cybercriminals. I care about this problem because technology is becoming a major part of daily life and people need to be safe online.

6. **How** would reality be different if this problem were solved? (This can be your opinion.)

If the problem were solved, individuals and organizations would have stronger cybersecurity systems to protect their information. People would feel safer using online services and businesses could operate more securely

Step 3: Understand and Quantify Your Problem

Next, you will conduct some basic web research to better understand, define, and quantify your problem. You will do this through a combination of Google search, Wikipedia, credible web sources, ChatGPT or other AI research tools, and your own synthesis of information from these sources. Be sure to give credit to your sources, and paraphrase (use your own words) rather than quoting directly.

7. What is the historical context for this problem? What happened in the past that contributes to the problem now?

Cybersecurity problems began increasing with the rapid growth of the internet and digital technologies. As more people started using online banking, social media, and digital services, cybercriminals also developed new ways to exploit systems and steal information. Weak security systems and lack of awareness about online safety have contributed to the increase in cybercrime over time.

8. What are the possible economic (money-related) reasons why this problem exists and continues?

Cybercrime is often motivated by financial gain. Hackers and cybercriminals steal money directly from bank accounts, conduct online fraud, or sell stolen personal data. Because cybercrime can generate large profits and sometimes has low risk of punishment, many criminals continue to engage in it.

9. What are the possible political reasons why this problem exists and continues?

Some countries have weak cybersecurity laws or limited enforcement, which allows cybercriminals to operate more easily. Governments may also struggle to keep up with fast-changing digital technologies, making it difficult to regulate cyber activities effectively.

10. What cultural beliefs and/or social norms possibly contribute to this problem?

Many people trust online messages, links, and emails without verifying their authenticity. Some users also reuse weak passwords or share personal information online without understanding the risks. This lack of cybersecurity awareness contributes to the problem.

11. Who are the people potentially responsible (directly or indirectly) for creating and/or maintaining this problem?

Cybercriminals and hackers are directly responsible for many attacks. Indirectly, organizations with weak security systems and individuals who do not follow safe online practices may contribute to the problem.

Now that you have the preliminary information you need, you'll continue your web research to find some numbers, or quantifiable information, to help describe your problem:

What numerical data can you find that is relevant to your problem? Be sure to use your own words and also cite (give credit to) your sources.

Example 1:

According to Chat GPT, there are about 880 white rhinos currently living in Kenya. This population is very small, and they are critically endangered.

Example 2:

Johannesburg has approximately 5.8 million residents (per ChatGPT) and had approximately 4.7 million international overnight visitors in 2019 (according to the South African Tourism Annual Report for 2019/2020).

12. Approximately how many people (and/or animals) are **directly** impacted by this problem? Explain.

Millions of internet users are directly affected by cybercrime each year. In Kenya alone, millions of people use mobile banking and internet services, which makes them potential targets for cyber attacks and online fraud.

13. Approximately how many people (and/or animals) are **indirectly** impacted by this problem? Explain.

Businesses, government institutions, and service providers are indirectly affected because cyber attacks can disrupt their systems, damage their reputation, and cause financial losses.

14. What other numerical data can you share that is relevant to your problem? What can you find out about its size and scope? What can be measured? (For example, the amount of trash produced in Nairobi each day, the number of people without access to clean water, etc.)

More than **5 billion people worldwide use the internet**, which increases the number of potential cybercrime targets. Reports also show that cyber attacks occur **every few seconds globally**, making cybersecurity a major global challenge.

Step 4: Describe Your Solved State

Without having to come up with *how* to solve the problem, describe what the desired, solved state looks like. Please use numbers wherever possible, and make your solved state-specific and measurable.

Example 1:

There would be a population of 10,000 healthy and protected white rhinos living in the wild in Kenya.

Example 2:

All 5.8 million residents of Johannesburg would have affordable and consistently available power from clean energy sources, 99.5% of the time.

15. If the problem were addressed/solved, what would reality be like?

All individuals, businesses, and government institutions in Kenya would have strong cybersecurity systems protecting their digital data 24/7. The number of successful cyberattacks would drop by at least **90%**, and online financial transactions would be secure. People would safely use mobile banking, e-commerce, and social media without fear of losing money or personal information.

16. Are there other benefits that would come from your problem being solved?
Name at least one.

1. Reduced financial losses from cybercrime.
2. Increased trust in online services and digital platforms.
3. Improved business growth and adoption of technology.
4. Fewer disruptions to government services and public infrastructure.

Step 5: Clarify Your Problem Scope

You are more effective at solving a problem when you know where its limits are. That is, when you know what is “in scope” and “out of scope.” For this reason, it is important to list what is out of scope, or NOT included as part of your problem definition.

Example 1:

The scope of the problem does not cover any other animal species besides white rhinos. It does not include white rhinos outside of Kenya.

Example 2:

The scope of the problem does not include any businesses or people outside of the legally-defined Johannesburg city limits. It does not apply to tourists or visitors staying for less than 1 year in Johannesburg.

17. What is NOT in the scope of your problem?

1. Cybersecurity threats outside Kenya.
2. Physical theft or offline financial fraud.
3. Cybersecurity issues caused by user negligence that is completely outside digital systems (like losing a password notebook).
4. Advanced international cyber warfare between countries.
5. Technology not connected to the internet (offline devices).

Step 6: Areas for Learning

What do you not know or understand that you would like to know more about? This can be anything related directly or indirectly to your problem. Let your curiosity run wild!

Example 1:

I'd like to know where most of the demand for rhino horn is coming from. I'd like to know who are the primary buyers and who is behind the trafficking of rhino horn. I'd like to know how long the average rhino's lifespan is. I'd like to know how many babies a typical female rhino has, and how many babies typically survive into adulthood. I'd like to know more about what diseases impact rhinos. I'd like to know more about the kinds of habitats that rhinos thrive in. I'd like to understand what international organizations do the best job supporting wildlife conservation and what their practices are. I'd like to know what models of community involvement have been most successful in keeping wildlife safe and thriving. I'd like to understand how much land is available in Kenya for rhinos to roam.

Example 2:

I'd like to better understand the utility company Eskom and its history. I'd like to understand why Eksom has failed to plan properly to update its infrastructure. I'd like to understand the relationship between Eksom and the South African government. I'd like to know if there are private utility companies providing competition to Eksom. I'd like to know what the latest breakthroughs are in solar power. I'd like to know what other possible energy sources might be made available in Johannesburg. I'd like to know how much energy tourists and temporary visitors use. I'd like to better understand the process of how limited energy supply gets allocated to people and businesses. I'd like to better understand the economic impact on people and businesses of not having power.

18. What else would you like to know or understand better? (It can be anything related to your problem.) List 5-10 things.

1. Which cybersecurity technologies are most effective in preventing online fraud?
2. How are hackers able to bypass existing security systems?
3. What are the top causes of data breaches in Kenya specifically?
4. How much money is lost to cybercrime each year in Kenya?
5. What are the best practices for individuals to stay safe online?
6. How do businesses protect themselves from cyber attacks?
7. How is the government regulating cybersecurity and enforcing laws?
8. What emerging cybersecurity threats should we anticipate in the next 5 years?
9. How do social engineering attacks work and how can they be prevented?
10. What are global examples of successful cybersecurity programs that could work in Kenya?

Step 7: Problem Statement

This step is the culmination of all you have done in Part A. You will synthesize the work you have done above to create a problem statement of 150 – 250 words. This should be in narrative form, 2-4 paragraphs, and should NOT use bullet points.

Your problem statement should:

- Provide a succinct description of the problem **in the first sentence**.

- Indicate the specific population affected
- Explain the impact (cost, time, environmental, personal) and why the problem matters.
- Explain what reality would be like if the problem were solved. The gap between present reality and the desired outcome should be clear.

Please cite (give credit to) where your information came from directly in your statement. Avoid word-for-word quoting and instead paraphrase (use your own words), as modeled in the example. Also, list your sources and their URLs (web addresses) at the end.

Example :

Kenya's white rhinos are in critical danger of extinction. There are currently about 880 white rhinos in the country of Kenya, per Wikipedia. According to Chat GPT, Rhinos are considered a keystone species, meaning they have a disproportionately large impact on their ecosystem compared to their population size. Rhinos help shape their environment by influencing vegetation growth and acting as seed dispersers, which creates habitat for other species (per ChatGPT).

The extinction of white rhinos would have cascading effects on other plant and animal species in their habitat. According to the Kenya Wildlife Service, rhinos' presence in reserves and parks brings millions of tourists each year, contributing to local economies and supporting conservation efforts. Once a species goes extinct, it is gone forever. The extinction of rhinos would represent the loss of millions of years of evolutionary history and unique genetic diversity that science has yet to fully understand and benefit from (per ChatGPT).

My problem would be considered solved when the population of wild, white rhinos in Kenya reaches 10,000, and when all imminent threats to their population including poaching and habitat destruction are not present. If this were the reality, it would create ecosystem balance, create large revenues from ecotourism, preserve

important cultural symbols, and allow for genetic diversity that could benefit humanity in ways we may not yet fully understand.

Sources:

Kenya Wildlife Services Annual Report 2017,
<https://www.kws.go.ke/content/annual-reports>

ChatGPT, <https://chat.openai.com/>

"White Rhinoceros", Wikipedia, https://en.wikipedia.org/wiki/White_rhinoceros

19. My problem is statement is:

Problem Statement Example for Cybersecurity in Kenya:

Kenya faces a growing cybersecurity problem, with thousands of cyberattacks occurring each year and millions of internet users at risk of fraud and data theft. Weak security systems, lack of awareness, and rapidly increasing digital adoption have made individuals and organizations vulnerable to hackers. Financial institutions, government systems, small businesses, and everyday citizens are all impacted, often resulting in stolen funds, compromised personal data, and a loss of trust in online platforms (Communications Authority of Kenya, 2023; ChatGPT, 2026).

The problem is particularly concerning for people who rely on mobile banking and e-commerce, as digital fraud and phishing attacks occur frequently. If these cyber threats continue unchecked, financial losses will increase, businesses may suffer reputational harm, and citizens may avoid using essential online services.

The solved state would see widespread implementation of strong cybersecurity measures, robust awareness programs for citizens, and effective government regulations. Online transactions would become secure, reducing successful

cyberattacks by at least 90%. Individuals would safely use digital platforms, businesses would operate without fear of breaches, and government systems would remain protected. This would strengthen trust in digital systems, encourage economic growth, and safeguard personal and organizational data, creating a safer online environment for millions of Kenyans (Cybersecurity Ventures, 2025).

20. Please list all sources that you used to form your problem statement:

- . Communications Authority of Kenya, 2023, <https://www.ca.go.ke/>
- . ChatGPT, 2026, <https://chat.openai.com/>
- . Cybersecurity Ventures, 2025, <https://cybersecurityventures.com/>



Please go back to Savanna and continue with your learning content. You will be prompted on when to return to complete Section B.

SECTION B: Research Questions & Hypothesis

IMPORTANT: Complete this section AFTER completing the Savanna Modules **Asking Effective Questions** and **conducting web research**.

Step 8: Research Questions

Based on what you have learned so far and on 'Step 6: Areas for Learning' from this worksheet, come up with 3 research questions. **Research questions should be complex enough that they can't be answered by a single Google search.** If appropriate, form a hypothesis that your research may confirm or reject. (As a reminder, a hypothesis is a prediction of how you think your research will answer your research question. It is your best guess. If you truly have absolutely no idea, state "not applicable.")

Example Research Question #1:

What are some ways can we increase rhino populations?

Hypothesis:

Rhino populations will increase by creating more open spaces for them to roam, increasing their protection, increasing international interest in them, and for other reasons, I have yet to uncover.

Example Research Question #2:

Which organizations have been effective at wildlife conservation and what practices do they use?

Hypothesis:

Not applicable; I don't know.

Example Research Question #3:

How many babies can a typical female white rhino have in her lifetime, and what are the reasons a female may not have high fertility?

Hypothesis:

A typical female white rhino can have 5 babies in her lifetime, and fertility may be affected by diet, amount of grazing territory, poaching, stress, mate availability, and other reasons I have yet to uncover.

21. Research question #1:

What are the main types of cyberattacks affecting individuals and businesses in Kenya, and how frequently do they occur?

Hypothesis (if applicable):

Phishing attacks, malware, and ransomware are the most frequent cyberattacks in Kenya, with small businesses and online banking users being most affected due to limited cybersecurity awareness and weak protective measures.

22. Research question #2:

Which cybersecurity practices and technologies are most effective in reducing cybercrime for both individuals and organizations in Kenya?

Hypothesis (if applicable):

Organizations and individuals using multi-factor authentication, firewalls, regular software updates, and employee cybersecurity training experience significantly fewer successful cyberattacks.

23. Research question #3:

How do government regulations, awareness campaigns, and public-private partnerships impact cybersecurity readiness and digital trust among Kenyan citizens?

Hypothesis (if applicable):

Stronger government regulations combined with public awareness campaigns and collaboration with private technology firms will lead to higher digital trust and reduce successful cyberattacks in Kenya.



Please go back to Savanna and continue with your learning content. You will be prompted on when to return to complete Section C.

PART C: Peer Activity Report

IMPORTANT: Complete this section AFTER completing the Savanna Lessons “Hub Activity: Problem Statement” and “Hub Activity: Web Research.”

Peer Activity #1: Problem Statement

Please report on your process of getting peer feedback.

24. Who reviewed your problem statement (item #19)? (Give the first and last names of your 2 peers.)

25. In brief, what feedback did they give to you?

26. Was their feedback useful to you? Did it feel kind? Why or why not?

27. Who did you give feedback to? (Give the first and last names of 2 peers—they may be the same or different peers from above.)

28. Do you feel that you gave useful and kind feedback to your peers? Why or why not?



After you updated earlier parts of the worksheet based on your feedback, please go back to Savanna and continue with your learning content. Return to Part E prior to submitting your milestone.

PART D: Daily 3 Challenge Report

Please answer the following questions honestly. There are no wrong answers! This is your opportunity for self-reflection.

29. How many days out of the past 7 did you do 20 minutes of movement?

30. How many days out of the past 7 did you write 3-morning pages?

31. Overall, how do you feel you are doing on building your Daily 3 habits?
Explain.

32. What is your biggest barrier to staying motivated?

33. If you have been practicing at least some movement and/or morning pages:

A) What effect (if any) have you noticed on your mood, focus, and productivity?

B) What helps you stay motivated?

34. If you haven't been practicing any movement and morning pages, what would motivate you to get started?

Once you have completed this worksheet:

1. Export/convert to .pdf.
 2. Rename it per the instructions.
 3. Upload to Savanna as your Milestone 3 Submission.
 - 4. Celebrate a job well done!**
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